

KAVALA MEGAS ALEXANDROS, Greece

WMO: 166240

Lat: 40.913N

Lon: 24.619E

Elev: 18

StdP: 14.69

Time Zone: 2.00 (EUE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	26.7	29.9	14.0	11.2	33.8	17.7	13.5	35.3	23.4	40.5	20.4	42.5	4.8	60	0.403

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	17.6	91.4	71.5	89.2	71.2	86.4	71.0	75.5	85.1	74.1	83.8	72.8	82.7	8.9	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
72.0	118.6	81.1	71.2	115.0	80.4	69.6	108.9	79.3	39.0	85.1	37.6	84.3	36.6	83.0	84.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
18.9	16.3	13.9	DB	21.6	95.6	3.9	2.7	18.8	97.5	16.6	99.1	14.4	100.6	11.6	102.6	
			WB	19.3	78.0	3.8	1.8	16.6	79.3	14.3	80.3	12.2	81.3	9.4	82.6	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	59.3	42.2	44.0	48.8	56.1	65.4	73.1	77.0	77.2	69.4	60.4	52.5	45.0
	DBStd	13.51	6.10	6.13	5.48	4.93	4.52	4.50	3.27	3.70	4.56	5.39	6.22	6.63
	HDD50	759	251	180	88	10	0	0	0	0	0	3	44	183
	HDD65	3290	707	588	503	270	52	2	0	0	15	157	376	620
	CDD50	4168	8	13	50	192	476	694	838	843	582	325	119	28
	CDD65	1221	0	0	0	1	63	245	373	378	147	14	0	0
	CDH74	10695	0	0	0	10	321	2006	3676	3686	942	54	0	0
	CDH80	3476	0	0	0	0	28	554	1312	1418	162	2	0	0
Wind	WSAvg	5.3	5.4	5.9	6.3	5.5	5.2	4.8	4.9	5.2	5.3	5.1	5.0	5.5
Precipitation	PrecAvg	25.0	2.8	2.5	2.3	1.6	1.7	1.3	1.0	0.7	1.4	2.4	3.0	4.2
	PrecMax	45.7	6.9	8.1	8.1	5.4	5.0	3.7	3.3	3.7	5.2	6.2	7.8	9.2
	PrecMin	13.6	0.0	0.3	0.2	0.1	0.6	0.3	0.2	0.0	0.0	0.1	0.7	1.1
	PrecStd	7.8	1.7	1.9	1.7	1.1	1.0	0.9	0.7	0.9	1.3	1.5	1.6	2.4
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	60.8	62.0	66.2	75.2	82.8	92.9	93.4	95.3	87.5	78.8	71.4	64.7
		MCWB	54.9	55.7	55.2	59.9	65.9	72.4	71.2	71.7	70.1	67.6	62.7	59.9
	2%	DB	57.4	59.0	62.8	70.2	80.0	88.1	90.0	91.7	84.0	75.2	68.0	60.9
		MCWB	52.5	53.1	55.5	58.5	65.5	71.2	71.2	71.5	69.1	65.0	61.6	56.8
	5%	DB	55.1	56.8	60.6	68.1	77.3	85.7	88.1	89.5	82.1	73.1	64.8	57.6
		MCWB	50.6	51.8	54.0	58.1	64.7	70.6	71.3	71.2	68.8	63.5	58.7	53.7
	10%	DB	51.9	53.8	58.6	65.9	75.3	82.8	86.2	87.4	79.4	70.1	62.5	55.3
		MCWB	47.9	49.5	52.6	56.6	63.7	69.4	71.2	71.2	67.2	61.7	57.4	51.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	56.7	56.9	58.4	63.2	69.7	76.3	76.6	78.1	73.6	70.0	64.4	60.5
		MCDB	59.8	60.5	63.1	72.1	78.8	86.2	86.6	86.6	81.6	75.8	69.4	63.4
	2%	WB	53.7	54.3	56.7	60.6	67.5	73.8	75.0	75.9	71.9	67.2	62.3	57.6
		MCDB	56.9	58.0	61.2	67.5	76.4	83.9	84.9	85.1	80.0	72.6	66.6	60.5
	5%	WB	51.2	52.2	54.8	59.2	66.2	72.2	73.7	74.5	70.3	65.1	60.3	54.8
		MCDB	54.1	55.8	59.3	65.9	74.4	82.4	83.5	84.1	78.5	70.7	64.1	57.4
	10%	WB	48.8	50.4	53.0	57.6	64.8	70.8	72.7	73.0	69.0	63.2	58.4	52.4
		MCDB	51.1	53.1	57.4	63.9	72.9	80.7	82.7	82.9	77.1	68.6	61.7	55.0
Mean Daily Temperature Range	5% DB	MDBR	13.0	13.7	14.1	15.5	16.1	17.0	17.6	18.7	17.4	16.0	14.7	13.3
		MCDBR	16.3	16.6	16.9	18.5	19.1	19.7	19.8	21.6	19.8	18.8	17.1	15.3
		MCWBR	12.2	12.0	11.3	10.1	9.2	9.0	9.0	9.6	10.6	11.0	11.7	11.8
	5% WB	MCDBR	14.3	15.0	15.7	16.6	17.3	18.1	18.0	19.2	18.0	17.2	15.2	12.8
		MCWBR	11.4	11.4	11.0	10.0	9.1	9.2	9.3	10.0	10.5	11.1	11.0	10.5
Clear-Sky Solar Irradiance	taub	0.356	0.388	0.435	0.488	0.471	0.467	0.464	0.472	0.454	0.417	0.387	0.366	
	taud	2.363	2.249	2.130	1.997	2.123	2.179	2.193	2.163	2.183	2.261	2.291	2.319	
	Ebn at Noon	248	256	256	250	258	259	258	252	247	243	233	232	
	Edh at Noon	27	35	44	54	49	46	45	45	42	35	29	27	
All-Sky Solar Radiation	RadAvg	550	816	1200	1582	1970	2155	2251	2021	1501	1004	653	489	
	RadStd	78	92	105	158	120	152	112	89	103	103	68	79	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (15 neighbors)	+0.83	N/A	N/A	N/A	N/A	N/A	-115	-263	+140	N/A

Nomenclature: See separate page